4. Domain: Cyber Security

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Rakshak DNS – DNS Threat Intelligence • AI-Powered Predictive Defence • Proactive Prevention of Access to Malicious Domains (Malware, Botnets, Phishing, etc.) • Enables near real-time analysis and response to sophisticated Multi-Vector Cyber Threats
	Name of the Project under which technology is developed	Centre of Excellence in DNS Security (CoEDNS)
	Funding Agency	NIXI
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	Rakshak DNS – Global & Enterprise DNS is an advanced Protective DNS solution designed to secure national-scale and enterprise networks against evolving cyber threats. Built with AI-powered predictive defence, Rakshak DNS enables proactive prevention of access to malicious domains associated with malware, botnets, phishing, and other attack vectors. Rakshak DNS supports near real-time analysis and has a high-fidelity threat intelligence with intelligent detection models. A key differentiator is its on-prem deployment capability, fine-tuned specifically for Indian DNS traffic patterns and threat landscapes, ensuring data sovereignty, low latency, and regulatory alignment. Rakshak DNS competes with leading global solutions such as Cisco Umbrella and Spamhaus, while offering localized intelligence, policy control, and deployment flexibility tailored to Indian scenario. Rakshak DNS is offered through: - Protective DNS as a Service (PDNS) for enterprises and government organizations Threat Feeds Subscription Service for SOCs, ISPs, and security vendors seeking high-quality, actionable DNS intelligence.
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Gopinath P, C-DAC Bangalore gopinath@cdac.in, 9845231819 Alternate email: rakshakdns@cdac.in
2.	**Brief details of Technology Developed**	Online Sucro-Crystal Imaging System (OSIS) is a machine vision system for Sugar industry.
	Name of the Project under which technology is developed	Collaboration Intelligent Transportation Systems Endeavour for Indian Cities' - InTranSE Phase II by CDAC-T, lIT-B, HT-M and BSc-Bangalore.
	Funding Agency	Ministry of Electronics and Information Technology
	Technology Readiness Level (TRL)	TRL 9
	Brief Write-up on the Technology Developed	OSIS facilitates automated measurement of crystal size for monitoring the growth of massecuite crystal during the crystallization pan process in the sugar industry. OSIS, the auto-sampling machine control system, captures massecuite from the pan, significantly enhancing real-time visual

Sr. No.	Parameters	Brief details
		inspection of the crystals during the crystallization pan process in the sugar industry
	Details of Contact Person along with Mobile No. and e-mail ID	Sreedhanya L. R, C-DAC Thiruvananthapuram 9847572121 sreedhanya@cdac.in
3.	**Brief details of the technology developed**	An endpoint security solution that allows whitelisting of USB storage devices on Windows Operating Systems.
	Name of the Project under which the technology is developed	USB Pratirodh
	Funding Agency	Internal Funding
	Technology Readiness Level (TRL)	TRL 8 and above
	Brief Write-up on the Technology Developed	The developed technology uses the Windows API and the File system mini-filter driver to ensure that only allowed USB storage devices can be accessed on the system. Link: https://www.cdac.in/index.aspx?id=print_page&print=cs_usb_pra Features 1. Device Control: All USB devices are uniquely identified. The user can add or remove devices to the database. The user can bind one or more USB devices for access using the enabled username. Any unauthorised new USB device cannot be accessed, unless it is registered. 2. User Authentication: Whenever a USB device is plugged in, the user is prompted to authenticate with a username and password. Only an authenticated user can access the device. If the user fails to authenticate, they receive an access denied message. 3. Secure Storage: Data on the USB storage devices can be encrypted. 4. Malware Detection: USB Pratirodh scans the plugged USB device for malware. Supported on: Windows 10 (64-bit).
	Details of the Contact Person along with Mobile No. and e-mail ID	Mr. Sandeep Romana, C-DAC Hyderabad 9959971727, sandeepr@cdac.in
4.	**Brief details of the technology developed**	An endpoint security solution that allows whitelisting of applications on Windows Operating Systems.
	Name of the Project under which the technology is developed	AppSamvid Project Name: Design and Development of an anti-malware solution for web applications and mobiles.
	Funding Agency	MeitY

Sr. No.	Parameters	Brief details
	Technology Readiness Level (TRL)	TRL 8 and above
	Brief Write-up on the Technology Developed	The developed technology uses the Windows API and the File system mini-filter driver to ensure that only whitelisted applications execute on the system. Link: https://cdac.in/index.aspx?id=cs_eps_appsamvid Features: 1. Whitelists executable and Java files (.exe, .class, .war, .jar). 2. Has Installation Mode: a. To allow the updating of software. b. To allow installation and/or uninstallation of software. 3. Folder Scan and File scan options to add executable files to the database. 4. Password-based access to the user interface. 5. Supports operating system updating via Microsoft Updates. 6. Bundled with a heuristic malware engine to gain confidence in which files to whitelist. 7. Allows files to be made as Trusted Updater. 8. Can identify potential updater files to help the user find which files can be made as trusted updater(s). Supported on: Windows 10 and 11 (64-bit).
	Details of the Contact Person alongwith Mobile No. and e-mail ID	Mr. Sandeep Romana, C-DAC Hyderabad 9959971727, sandeepr@cdac.in
5.	**Brief details of Technology Developed**	Hardened Android Platform
	Name of the Project under which technology is developed	Meghdooth Project Name: "Deployment of Secure mobile Email, Voice, SMS and Mobile Device management solution for Govt. Users (Meghdoot)"
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 8
	Brief Write-up on the Technology Developed	Major features supported include ● Removal of unwanted Stock Applications ● Elevation of Permission security protection levels to restrict permissions only to system applications ● App installation control with Whitelisted App installation support ● Developer option protection ● Resource Control (Camera, Internet, Bluetooth, Location) to Privileged Applications Devices Supported : Pixel 7a (Android 13)

Sr. No.	Parameters	Brief details
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr Chaithanya M K, C-DAC Hyderabad mkchaithanya@cdac.in, +91 91604 41112
6.	**Brief details of Technology Developed**	Abhedya - SecureApp SDK A security SDK providing plug-and-play mobile security solution that enables mobile app developers for easy integration of security features
	Name of the Project under which technology is developed	Project Name: Design and Development of Advanced Mobile Security and Analysis Solutions
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 8
	Brief Write-up on the Technology Developed	The SDK is a plug-and-play mobile security solution that enables developers to easily integrate essential security features into their mobile applications. It provides runtime protection including ● Device Environment Check ● Device Misconfigurations Detection ● Secure Communication Configurations Checks ● Instrumentation Frameworks Detection These features help applications remain secure against modern mobile threats while reducing development complexity. The exact features supported may vary depending on the underlying platform i.e. Android, iOS & cross-platform frameworks.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr Chaithanya M K, C-DAC Hyderabad mkchaithanya@cdac.in, +91 91604 41112
7.	**Brief details of Technology Developed**	InTrust tracks assets, services, and vulnerabilities to continuously assess and derive the overall security posture of the organization or network.
	Name of the Project under which technology is developed	Zero Trust Network Access
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	8
	Brief Write-up on the Technology Developed	InTrust – Asset & Traffic Monitoring for ZTNA is a security solution designed to provide continuous visibility and trust assessment for Zero Trust Network Access environments. It tracks assets, services, and vulnerabilities to derive an accurate and evolving security posture, enabling informed access and policy decisions. A key differentiator of InTrust is its agent-less, non-intrusive architecture, allowing seamless deployment without impacting existing systems or network performance. The platform supports near real-time security evaluation by automatically discovering assets, monitoring traffic, and dynamically calculating trust scores based on observed behavior and risk indicators.

Sr. No.	Parameters	Brief details
	Details of Contact Person along with Mobile No. and e-mail ID	Dr. Muraleedharan N, C-DAC Bangalore murali@cdac.in, 7795565027
8.	**Brief details of Technology Developed**	XShield360 – From Detection to Response, Powered by Intelligent Automation. It functions as an extended detection, security orchestration, automation, and Response system, enabling coordinated threat detection, automated workflows, and rapid incident response across the security stack.
	Name of the Project under which technology is developed	-
	Funding Agency	-
	Technology Readiness Level (TRL)	8
	Brief Write-up on the Technology Developed	XShield360 offers strategic autonomy and flexibility compared to global SOAR and XDR platforms. Its native integration with both open-source and security tools makes it especially suitable for government and regulated environments. The platform emphasizes automation and orchestration through configurable playbooks for IT threat response, reducing manual effort and response time. Key capabilities include: - Automated risk assessment and prioritization - AI-based alerts, incident and case management, incident triage, and escalation - Threat intelligence management, including enrichment and correlation - Comprehensive dashboards and reporting with real-time performance metrics, incident status, and vulnerability insights Role-Based Access Control (RBAC) for secure, policy-driven operations
	Details of Contact Person along with Mobile No. and e-mail ID	Mr. Praveen Ampatt, apraveen@cdac.in, 9448429212 Ms. Surabhi Diwedi, surabhi@cdac.in, 9969287560 C-DAC Bangalore
9.	**Brief details of Technology Developed**	Secure Vault Solution for secure storage, protection, and management of Personally Identifiable Information (PII) such as Aadhaar, PAN, Voter ID, Driving License, Passport, etc., with HSM-based encryption and regulatory compliance
	Name of the Project under which technology is developed	Secure Vault Solution / Aadhaar Data Vault (ADV)
	Funding Agency	Secure Vault solution is offered as a product/solution to entities willing to securely store and manage secure PII data in its own infrastructure.
	Technology Readiness Level (TRL)	TRL 7 and above (Production-grade, nationally deployed)
	Brief Write-up on the Technology Developed	Secure Vault is a production-grade, indigenous cybersecurity solution developed by C-DAC Mumbai for the secure storage and management of Personally Identifiable Information (PII) such as Aadhaar, PAN, Voter ID, Driving License, passport details, and associated PII data. The solution is already deployed by major entities such as the Income Tax Department, Employees' Provident Fund Organisation (EPFO), WBEDIC, and Tripura State Cooperative Bank.

Sr. No.	Parameters	Brief details
		It is offered as a customizable product to government, PSU, and private entities to meet regulatory compliance such as UIDAI guidelines and the Digital Personal Data Protection Act (DPDPA). The technology addresses critical security and privacy risks arising from large-scale digital usage of PII across government, finance, healthcare, and public service ecosystems. Secure Vault provides a centralized and encrypted repository with strong access controls, preventing unauthorized access, misuse, and data breaches while maintaining high performance. Secure Vault implements a multi-layered security architecture using advanced cryptographic algorithms such as AES, RSA, ECC, and hybrid encryption, with support for both software-based and Hardware Security Module (HSM)-based cryptographic key management. HSM integration ensures tamper-resistant protection of encryption keys and adherence to global best practices. The solution supports end-to-end encryption for data at rest and in transit, role-based access control (RBAC), secure API-based access, and comprehensive audit logging and monitoring. Key Features: • Centralized, secure storage for sensitive PII data • End-to-end encryption (data at rest and in motion) • HSM-based and software-based cryptographic key management • Support for AES, RSA, ECC, and hybrid encryption mechanisms • Strict access control with detailed audit trails and monitoring • High throughput and scalability for large-scale deployments • Flexibility and customization for Government, PSU, and private organizations • Compliance with regulatory requirements related to PII data, including UIDAI guidelines and the Digital Personal Data Protection (DPDP) Act. Secure Vault provides a trusted, scalable, and compliant platform for secure PII management, strengthening data privacy, citizen trust, and India's digital infrastructure.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Name: Mr. Vijay Jain, C-DAC Mumbai Email Id: vijayj@cdac.in Contact Number: 9833012311
10.	**Brief details of Technology Developed**	Bump-In-The-Wire (BiTW) solution for SCADA Security.
	Name of the Project under which technology is developed	Development of Security Solutions for SCADA Systems.
	Funding Agency	Ministry of Electronics and Information Technology (MeitY), Government of India
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	The Bump-In-The-Wire (BiTW) solution is designed and implemented to improve the security of SCADA systems against cyber-attacks, with it's easy placement option without making changes in the end devices, including the legacy devices. It allows easy and seamless integration into an existing system to enable security, using C-DAC's developed protocol, to secure the communication between remote terminal unit (RTU) and master terminal units (MTU).

Sr. No.	Parameters	Brief details
		The solution is for IEC 60870-5-101 protocol in transmission power SCADA with data link layer security to significantly enhance protection against the cyber attacks.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Anupam Saxena, C-DAC Mumbai Mob. No,: +91 9920817771 e-mail: anupam@cdac.in
11.	**Brief details of Technology Developed**	Secure Vega Transfer
	Name of the Project under which technology is developed	Project Name: POC towards Establishment of Secure Distribute Data Grid for High Speed Data Transfers between HPC System
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 5 (Current) TRL 8 (planned April 2026)
	Brief Write-up on the Technology Developed	Traditional TCP transfers can't keep up with today's data demands. Our solution bypasses the bottlenecks of latency and congestion by utilizing optimized UDP-based technology, delivering high-speed, secure, and reliable data movement across any network USPs ● Optimized UDP protocol for rapid, high-volume Transfers of multi-terabyte files over High-Speed Networks or WAN ● Support User authentication and secure communication Features and Enhancements Roadmap ● Supports schedule File Transfers ● Auto resume Supports auditing and reporting features.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr Sai Gopal T, C-DAC Hyderabad saigopalt@cdac.in, +91 8143532340